

Deep Learning for Solving Dynamic Models

Session 7: Structural estimation via simulated method of moments

Simon Scheidegger

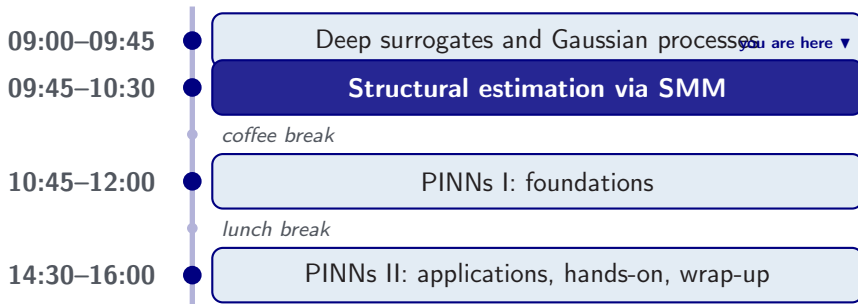
HEC, University of Lausanne

Based on Chen, Didisheim & Scheidegger (2026), *Journal of Financial Economics* 177, 104222

University of Padova

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Day 2 at a Glance



Roadmap of This Lecture

I. The idea

- ▶ The estimation loop, and why it is expensive
- ▶ SMM on one frame; common random numbers
- ▶ The model and the moments

II. One parameter

- ▶ A policy network with ϱ as an input
- ▶ Its accuracy against the reference solution
- ▶ Moments, criterion, estimate (07_01)
- ▶ The other route: a GP of the moment map

III. Two parameters

- ▶ (β, ϱ) jointly (07_02)
- ▶ How sharply each moment set pins down β
- ▶ The moment Jacobian in one number
- ▶ The GP route with two parameters

Finger exercise, production, hands-on

Notebooks in `day2/code/07_structural_estimation/`: `07_01_SMM_One_Parameter.ipynb`,
`07_02_SMM_Two_Parameters.ipynb`

Learning Objectives

By the end of this lecture you should be able to:

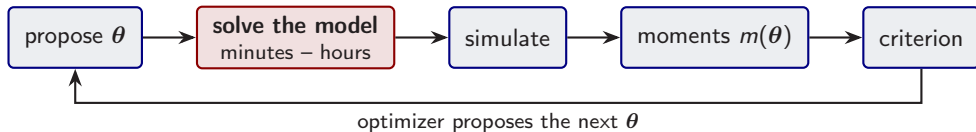
- ▶ Write down the simulated-method-of-moments criterion and say why every evaluation of it used to require solving the model
- ▶ Train a policy network with structural parameters as inputs, and validate it against a reference solution across the parameter range
- ▶ Simulate moments under common random numbers and minimize the criterion on the fly, for one parameter and for two, and do the same with Gaussian processes of the moment map fit to a handful of solves
- ▶ Read a criterion surface: say how sharply a moment set pins down each parameter, and confirm it with the moment Jacobian

Part I

The Idea

The Estimation Loop, and Why It Is Expensive

The nested loop. For each candidate θ :



The surrogate move. Train the policy network with θ as an input, once. The red box is then gone from the loop: a candidate θ costs one simulation of the network, and the loop runs on the fly.

Two consequences

The criterion is cheap enough to evaluate on a grid, so you can *look at* it rather than trust a local optimizer's report; and under common random numbers it is smooth in θ , so a standard minimizer finishes the job.

SMM on One Frame

Data: T observations $\{y_t\}$, summarized by q moments $\hat{m} = \frac{1}{T} \sum_t g(y_t)$: here standard deviations, autocorrelations, a mean.

Model: for a candidate θ , simulate a sample of length S and compute the same moments, $m(\theta)$.

The estimator

$$\hat{\theta} = \arg \min_{\theta} Q(\theta), \quad Q(\theta) = [m(\theta) - \hat{m}]^T W [m(\theta) - \hat{m}], \quad W = I \text{ here.}$$

- ▶ $q \geq p$ moments for p parameters; with $q > p$ the criterion cannot reach zero and its minimum is a compromise across moments.
- ▶ $W = I$ is consistent and simple; the efficient choice, the inverse covariance of the data moments, changes the standard errors, not the picture.
- ▶ **Identification:** θ_0 is the unique minimizer of the population criterion, so the moments must *move* with every parameter (Part III).

Common random numbers

Use the **same** shock sequence for every candidate θ . Then $m(\theta)$ is a smooth function and Q has a clean minimum. Redraw the shocks and the minimizer chases simulation noise.

The Model: Brock–Mirman with Stochastic TFP

The economy of Session 2, with $\delta = 0.10$ so that no closed form exists and the Euler equation has to be solved:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \quad K_{t+1} + C_t = Y_t + (1 - \delta)K_t,$$

$$Y_t = z_t K_t^\alpha, \quad \log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

Calibration

α	δ	σ	β (Part II)
0.36	0.10	0.04	0.96

Truth for the synthetic data: $\varrho = 0.90$,
and in Part III $\beta = 0.96$.

Four moments, and which parameter each responds to

moment	in ϱ	in β
Std($\Delta \log C_t$)	rises above 0.8	weak
Corr($\Delta \log C_t, \Delta \log C_{t-1}$)	rises to 0.9, falls after	weak
Corr($\log Y_t, \log Y_{t-1}$)	rises throughout	none
mean(l_t/Y_t)	none	rises

ϱ sets how long shocks last, so the autocorrelations respond to it. β sets the steady-state capital stock,
 $k^* = (\alpha/(1/\beta - 1 + \delta))^{1/(1-\alpha)}$, so only a level moment responds to it. Measured in 07_01 and 07_02.

The Reference Solution, and the Data

Reference solver (`brock_mirman_reference.py`): iterate on the Euler equation with cash on hand as the state (endogenous grid method, cubic splines) until the consumption function stops changing. Relative Euler errors below 10^{-6} on the ergodic set at every corner of the parameter box; about 1.7 s per solve.

The data: one path of $T = 500$ periods from the reference solution at the true parameters, with its own shocks. Its four moments are the targets $\hat{m}$.

What the reference solver is for

Not for the estimation, where it would be the expensive red box. For **validating the surrogate**, at several parameter values, on the states the model visits; and, once, for drawing the criterion the true model would give, so that the surrogate's criterion can be laid on top of it.

Part II

One Parameter: ϱ

Step 1: A Policy Network with ϱ as an Input

The surrogate

$(\log z_t, K_t, \varrho) \mapsto$ consumption share of cash on hand $m_t = Y_t + (1 - \delta)K_t$, through a sigmoid, so $0 < C_t < m_t$ by construction and $K_{t+1} = m_t - C_t$.

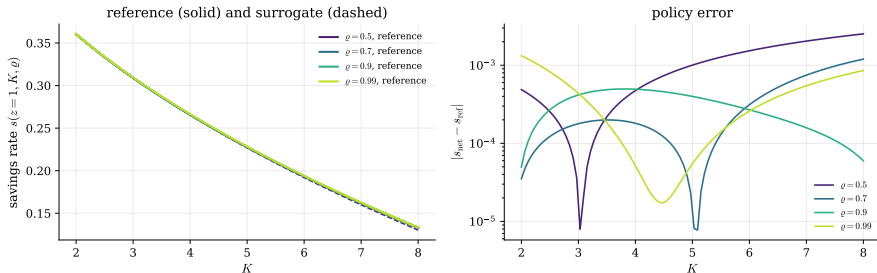
Loss: the relative Euler residual, with five Gauss–Hermite nodes for the expectation,

$$e(z, K, \varrho) = 1 - \frac{1}{\beta C_t \mathbb{E}_t \left[\frac{1 - \delta + \alpha z_{t+1} K_{t+1}^{\alpha-1}}{C_{t+1}} \right]}, \quad \mathcal{L} = \frac{1}{N} \sum_i e_i^2.$$

States: each batch is half uniform draws over a box in $(\log z, K, \varrho)$, half states visited by simulating the current policy, so the network is accurate where the estimator will evaluate it.

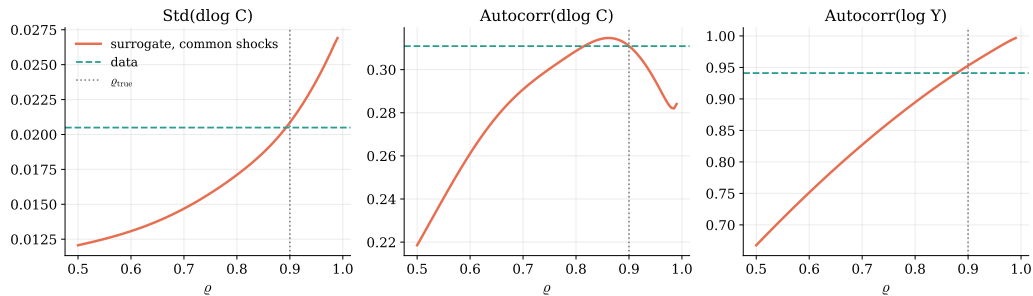
Nothing about ϱ is special. It enters the transition $\log z_{t+1} = \varrho \log z_t + \sigma \varepsilon$ inside the expectation, and the Euler residual is zero for every value of it. One training run, 2 min, covers $\varrho \in [0.50, 0.99]$.

Step 1, Checked Against the Reference Solution



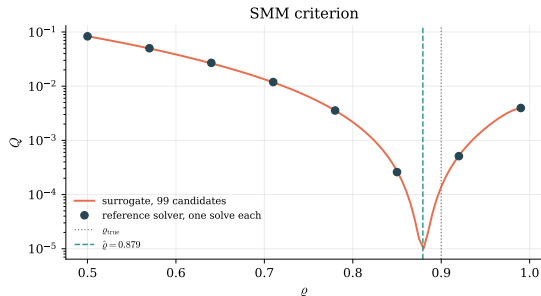
Notebook 07_01, stored teaching run. On 2,000 ergodic states per ϱ : $\varrho = 0.5$: mean Euler error $1.6 \cdot 10^{-4}$, mean consumption error $8.3 \cdot 10^{-4}$, max $1.3 \cdot 10^{-3}$; $\varrho = 0.7$: mean Euler error $5.5 \cdot 10^{-5}$, mean consumption error $2.1 \cdot 10^{-4}$, max $4.9 \cdot 10^{-4}$; $\varrho = 0.9$: mean Euler error $1.1 \cdot 10^{-4}$, mean consumption error $5.7 \cdot 10^{-4}$, max $7.1 \cdot 10^{-4}$; $\varrho = 0.99$: mean Euler error $2.5 \cdot 10^{-4}$, mean consumption error $1.1 \cdot 10^{-3}$, max $2.1 \cdot 10^{-2}$.. The consumption error is the number that matters for estimation; the Euler residual is only the training loss.

Step 2: ϱ Moves the Simulated Moments



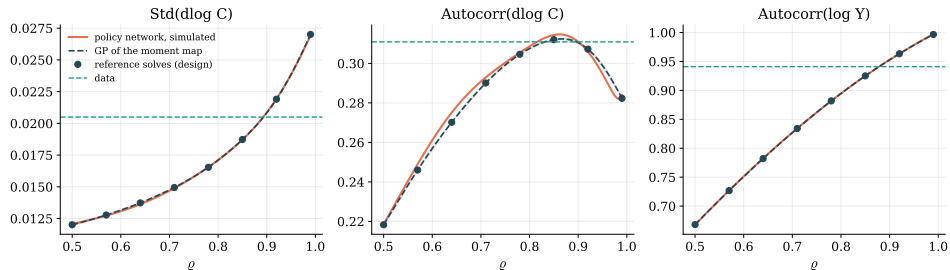
99 candidates from 0.50 to 0.99, each simulated for 2,000 periods from the same state under the **same** shocks, in 1 s. Every moment used moves with ϱ , so the data moments (dashed) pin it down; the mean investment share does not move with ϱ and is left out. Notebook 07_01, stored teaching run.

Step 3: The Criterion and the Estimate



Notebook 07_01. The surrogate's criterion on 99 candidates, and the reference solver's on 8 candidates, one solve each. Grid minimum, then a bounded scalar minimizer: $\hat{\rho} = 0.8793$. **The surrogate's curve lies on the reference solver's points:** the surrogate adds no visible error. **The minimum is at 0.8793, not 0.90, for both:** that is the sampling error of a 500-period sample (its output autocorrelation is 0.941 against a population value of 0.947; across 200 such samples the sampling standard deviation is 0.0127).

The Other Route: a GP of the Moment Map



The network is a surrogate of the *policy*, simulated at every candidate. The other kind of surrogate from Session 6 fits the *moment map* $m_j(\varrho)$ directly: one GP per moment on 8 reference solves (dots), Matérn 5/2, hyperparameters by marginal likelihood. Its posterior mean replaces the simulation in Q ; the band is its own error bar. Estimate: 0.8784 by the GP route against 0.8793 by the network route, truth 0.90. Notebook 07_01, Section 6.

Part III

Two Parameters: (β, ϱ)

The Same Surrogate, One More Input

$(\log z_t, K_t, \beta, \varrho) \mapsto$ consumption share; the Euler residual with β read from the input. Box $\beta \in [0.92, 0.99]$, $\varrho \in [0.50, 0.99]$; the capital box is wide enough for every β 's ergodic set.

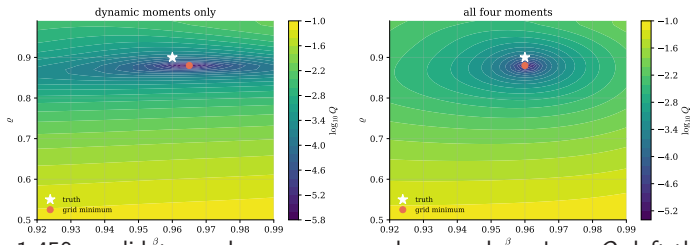
Checked at five corners and the centre of the box, on 2,000 ergodic states each:

β	ϱ	mean Euler	max Euler	mean $ \Delta C /C$	max $ \Delta C /C$
0.92	0.5	$1.3 \cdot 10^{-4}$	$3.5 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	$7.6 \cdot 10^{-4}$
0.92	0.99	$1.8 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$	$5.9 \cdot 10^{-4}$	$3.0 \cdot 10^{-3}$
0.96	0.9	$3.1 \cdot 10^{-5}$	$1.9 \cdot 10^{-4}$	$9.8 \cdot 10^{-5}$	$5.3 \cdot 10^{-4}$
0.99	0.5	$3.0 \cdot 10^{-4}$	$4.4 \cdot 10^{-4}$	$2.3 \cdot 10^{-3}$	$2.9 \cdot 10^{-3}$
0.99	0.99	$5.3 \cdot 10^{-4}$	$5.1 \cdot 10^{-3}$	$3.7 \cdot 10^{-3}$	$3.3 \cdot 10^{-2}$

Why the check comes first

A criterion that is flat in β has two possible causes: the moments do not respond to β , or the *surrogate* does not. Only the comparison with the reference solution tells them apart. The table rules the second out here.

How Sharply Each Moment Set Pins Down β



Floor of each surface,
 $\min_{\varrho} Q(\beta, \varrho)$:

β	dynamic only	all four
0.920	$3.4 \cdot 10^{-4}$	$4.2 \cdot 10^{-3}$
0.960	$5.6 \cdot 10^{-6}$	$6.2 \cdot 10^{-6}$
0.990	$1.4 \cdot 10^{-4}$	$5.3 \cdot 10^{-3}$

1,450 candidates under common random numbers, $\log_{10} Q$; left the three dynamic moments, right with the mean investment share added. Both find the same minimum. Moving β from 0.96 to the edge of the box raises the criterion by a factor 25 to 60 with the dynamic moments, by 700 to 850 with the level moment added: **the dynamic moments identify β , but 12 to 37 times more weakly**, so a small error in the moments moves $\hat{\beta}$ far along the left surface and little along the right one. **Identification is fixed by information, not by a better optimizer.** Notebook 07_02.

The Moment Jacobian in One Number

At the estimate, by central differences on the surrogate, with the parameters scaled to the box, $u = (\theta - \theta_{lo})/(\theta_{hi} - \theta_{lo})$, so that a step of one is a step across the plotted range in either direction (in raw units the β column would be seven times smaller only because the box is seven times narrower):

$$M = \frac{\partial m}{\partial u} \in \mathbb{R}^{q \times 2}, \quad M = U \text{diag}(s_1, s_2) V^\top.$$

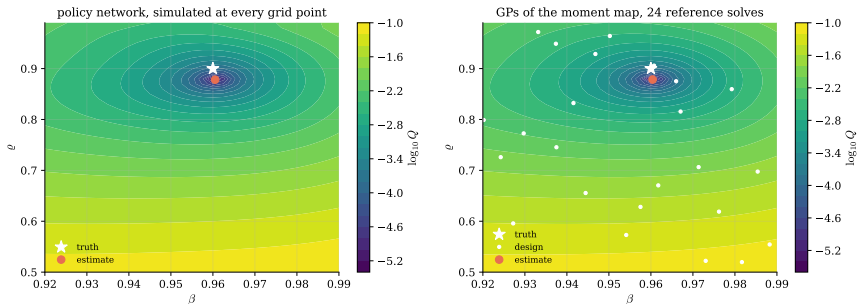
Near the minimum $Q \approx \delta u^\top M^\top M \delta u$, so s_2^2 is the criterion's curvature in its flattest direction and the last column of V names that direction. This is the previous frame in one number.

moments	s_1	s_2	flattest direction (β, ϱ)
dynamic only	0.273	0.0320	$(-1.00, -0.01)$
all four	0.272	0.1419	$(-1.00, -0.02)$

Estimates: dynamic moments only $(\hat{\beta}, \hat{\varrho}) = (0.9641, 0.8786)$; all four moments $(0.9605, 0.8782)$; truth $(0.96, 0.90)$. The ratio of singular values falls from 9 to 2 when the level moment is added.

With an efficient W and the covariance of the data moments, $(M^\top W M)^{-1}$ scaled by $1/T$ is the asymptotic variance of $\hat{u}$; a small s_2 is a large standard error along V_2 . The notebooks stop at M ; the covariance is one more simulation away.

The GP Route With Two Parameters



Left: the criterion from the policy network, simulated at every grid point. Right: the criterion from four GPs of the moment map fit to 24 reference solves on a Latin-hypercube design (dots). Estimates (0.9604, 0.8785) by the GP route, (0.9605, 0.8782) by the network route, truth (0.96, 0.90). The two routes agree; what differs is the cost curve: the network adds one input per parameter and trains once, the GP needs a design that fills one more dimension per parameter and carries an error bar for it. Notebook 07_02, Section 6.

The Surrogate Is Not Free

The moments came from the network, not from the model: $m_{\text{net}}(\theta) = m(\theta) + e(\theta)$.

e does not shrink with the sample

More data makes $\hat{m}$ precise; it does nothing to e . The estimator converges to the minimizer of the criterion built on m_{net} , which is the truth only to the extent that e is small *across the parameter range*.

The practical rule. Report the surrogate's **maximum** consumption error over the parameter box against a reference solution, on the states the model visits, before reporting an estimate. In 07_01 that maximum is $2.1 \cdot 10^{-2}$; in 07_02 it is $3.3 \cdot 10^{-2}$. If it is not small relative to how much the moments move, train longer or widen the state box, and check again.

Chen, Didisheim & Scheidegger (2026), fn. 15, make the same point about their own surrogate.

Finger Exercise: Which Moment Moves With ϱ ?

PAUSE: 5 min

The setting

$\log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}$ with $\varepsilon \sim \mathcal{N}(0, 1)$, stationary.

1. Compute $\text{Var}(\log z_t)$ and $\text{Var}(\Delta \log z_t)$ as functions of ϱ and σ .
2. Which of the two rises with ϱ , and which falls?
3. Output is $Y_t = z_t K_t^\alpha$ and capital moves slowly. Which of the four moments on the model frame should respond most to ϱ , and in which direction? Check against the moment figure of Part II.

Solution: Which Moment Moves With ϱ ?

1. The two variances

Stationarity gives $\text{Var}(\log z) = \sigma^2 / (1 - \varrho^2)$. For the difference,

$$\Delta \log z_t = (\varrho - 1) \log z_{t-1} + \sigma \varepsilon_t, \text{ so } \text{Var}(\Delta \log z) = (1 - \varrho)^2 \frac{\sigma^2}{1 - \varrho^2} + \sigma^2 = \frac{2\sigma^2}{1 + \varrho}.$$

2. Directions

The level variance **rises** without bound as $\varrho \rightarrow 1$; the variance of the difference **falls**, from $2\sigma^2$ at $\varrho = 0$ to σ^2 at $\varrho = 1$. A more persistent process wanders further but changes less from one period to the next.

3. The moments

$\text{Corr}(\log Y_t, \log Y_{t-1})$ inherits the autocorrelation of $\log z$, close to ϱ itself, and rises steeply. $\text{Std}(\Delta \log C)$ follows the falling variance of the difference, dampened by consumption smoothing, so it moves little. The mean investment share depends on the level of capital, which ϱ hardly touches. That is the ordering in the Part II figure.

From Classroom to Production

Brock–Mirman is the mechanism in miniature. Chen, Didisheim & Scheidegger (2026) run it on a high-dimensional option-pricing model, with a network in place of a Fourier-transform pricer.

What the surrogate enables

- ▶ Re-estimate **every day** instead of once per paper
- ▶ Systematic out-of-sample evaluation against reduced-form benchmarks
- ▶ Conditional distributions of option returns

What that produced

- ▶ An option-implied **tail-risk index**, predictive of market crashes
- ▶ A **parameter-instability** measure that lines up with option-market illiquidity
- ▶ A read on the structural-versus-reduced-form trade-off

The pattern worth remembering

None of these are faster versions of old results. They are questions whose *answers require a time series of estimates*, and that is what a surrogate with the parameters as inputs manufactures.

Summary, and What to Run

1. The expensive step in SMM is the model solve inside the loop. A policy network with the parameters as inputs is trained once and removes it.
2. Validate the surrogate against a reference solution across the parameter range before estimating with it; the Euler residual is not that validation.
3. Common random numbers make the simulated moments smooth; then a grid shows the criterion and a standard minimizer finishes it.
4. Identification is a property of the moments. A criterion that is flat along a parameter is either the moments or the surrogate; the reference check tells which, and the moment Jacobian puts a number on it.
5. Two surrogates, one estimate: a policy network simulated on the fly, or Gaussian processes of the moment map fit to a few solves. Both replace the solve in the loop; the label budget and the number of parameters decide between them.

Hands-on, day2/code/07_structural_estimation/

07_01_SMM_One_Parameter.ipynb (3 min) and 07_02_SMM_Two_Parameters.ipynb (8 min), teaching presets stored; set `RUN_MODE = "smoke"` in class. Try: change the true ϱ ; drop a moment; shorten the observed sample to $T = 100$ and watch the criterion's minimum move.



Questions?

Simon Scheidegger

HEC, University of Lausanne

simon.scheidegger@unil.ch · sischei.github.io

Next up, Session 8: *Economics-informed neural networks: foundations.*

Materials:

github.com/sischei/deep_learning_padova_0926